

Fine-Tuning Pipeline

Llama-3.1-8B-Instruct
(pre-trained, FP16)

QLoRA Fine-Tuning
83K cyber examples
rank-16, 2 epochs

Fine-Tuned Checkpoint (FP16)
WMDP 44--46% | MMLU 63--65%

PTQ

C1 Generic
(WikiText-2)

C2 Self-gen
(T=0.8)

C3 DACS
(domain-aligned)

3 × 3 Calibration × Format Matrix (cyber MMLU %)

FP8 E4M3

AWQ INT4

SQ INT8

62.18%

59.42%

59.00%

62.18%

59.72%

49.92% (!)

62.18%

59.64%

56.22%

eval

Evaluation

WMDP-Cyber
1,987 MCQs

MMLU
14,042 MCQs
(5-shot)

Self-cal failure
0.9.08 pp